Student's Index No.: _____ Signature: _____

UNIVERSITY OF GHANA

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DEPARTMENT OF TEACHER EDUCATION

SCHOOL OF EDUCATION AND LEADERSHIP

COLLEGES OF EDUCATION

END OF SEMESTER TWO EXAMINATIONS, 2024/2025

TIME: 2 HOURS

COURSE CODE: **TEUP 208**

COURSE TITLE: **INTEGRATED SCIENCE II FOR UPPER PRIMARY**

INSTRUCTIONS: This paper consists of **Sections A and B**. Answer all the questions in Section A and three from Section B.

SECTION A

[25 marks]

Answer all the questions in this section.

1. Energy resources derived from natural organic materials are called _____.
   A. Geothermal energy source B. Fossil fuels C. Biomass D. Natural gas

2. Which among the following causes day and night?
   A. Gravity B. Rotation . Revolution D. Orbiting

3. Which of the following is a viral skin disease?
   A. Acne B. Candidiasis C. Chicken pox D. Ring worm

4. Given the relation $KE = \frac{1}{2}mv^2$, where the symbols represent their usual meanings, calculate the kinetic energy of a body of mass 1500Kg at rest.
   A. 750J B. 0J C. 1500J D. 3000J

5. The natural forest is _____.
   A. a non-renewable energy resource
   B. renewable energy resources
   C. geothermal energy resources
   D. hydrological energy resource

6. Which of the following is not a contact force?
   A. Cohesive force B. Compression C. Frictional Force D. Gravitational Force

7. Which of the following is not an effect of force?
   A. A force can change the shape of an object.
   B. A force can cause a change in the direction of a moving object.
   C. A force can cause a moving object to accelerate or to decelerate.
   D. A force cannot cause a change in momentum which is the product of a body's mass and its velocity

8. Which is the weakest force in nature?
   A. Cohesive force B. Compression C. Frictional Force D. Gravitational Force

9. What is the causative organism for Measles?
   A. human papilloma virus
   B. rubeola virus
   C. treponema pallidum
   D. varicella-zoster virus

10. The Earth's only proper natural satellite is the _____.
   A. Europa B. Moon C. Sun D. Titania

11. In a hydroelectric dam, what type of energy is converted into electrical energy?
A. Solar energy
B. Mechanical energy
C. Thermal energy
D. Nuclear energy

12. Which of the following is NOT a common use of solar energy?
A. Heating water in solar water heaters
B. Generating electricity with solar panels
C. Powering cars with solar cells
D. Growing plants in greenhouses

13. What is the primary source of energy for photovoltaic (PV) solar panels?
A. Wind
B. Natural gas
C. Sunlight
D. Geothermal heat

14. What is a satellite in the context of space technology?
A. A celestial body orbiting a star
B. A natural moon orbiting a planet
C. A man-made object placed into orbit around a celestial body
D. A type of meteorite

15. What happens to the solar energy absorbed by plants during photosynthesis?
A. It is converted into mechanical energy
B. It is stored as chemical energy in the form of glucose
C. It is released as thermal energy
D. It is converted into nuclear energy

16. Which of the following is not a renewable source of energy?
A. Wind
B. Solar
C. Nuclear
D. Water

17. The following are examples of trace fossils **except** _____
A. Footprints
B. Burrows
C. Bones
D. Nests

18. Which of the following is a contact force?
(A) Electrostatic force (B) Electromagnetic (C) Magnetic force (D) Frictional force

19. The following are layers of the skin except
A. Dermatitis
B. Epidermis
C. Dermis
D. Hypodermis

20. The following are examples of non-renewable sources of energy except
(A) Dry cells (B) Nuclear energy (C) Fossil fuel (D) Biomass

21. Fast moving vehicles overturn when negotiating curves because of _____.

22. A plastic comb passed through dry hair a few times can pick up pieces of paper because _____.

23. The core competency that enables learners to draw on their own experiences to analyse situations and choose the most appropriate solution from a few possible options is known as _____.

24. The best method or technique used normally for separation of a mixture of different compounds in solution is _____.

25. The _____ is organic material that comes from plants and animals, and includes crops, waste wood, and trees.

SECTION B

[75 marks]

Answer three (3) questions from this section.

1.

a. Describe briefly how a mixture of granulated sugar and sand can be separated. **(5 marks)**

b. i) What is force as applied in science? **(2 marks)**

ii) Explain briefly why it is easier to break an egg striking its flat side rather than the pointed edge. **(5 marks)**

c. i) Differentiate between adhesion and cohesion forces. **(4 marks)**

ii) Differentiate between centrifugal force and centripetal force. **(4 marks)**

d. Write down **FIVE** uses of solar energy to mankind. **(5 marks)**

2.

a. Choose one of the mentioned skin diseases (ringworm, eczema, chickenpox, and measles) and provide a detailed description of its causes, symptoms, and common treatments. **[9 marks]**

b. Explain the fundamental difference between luminous and non-luminous bodies in space. Provide examples of each. **[6 marks]**

c. Discuss the role of personal hygiene in preventing skin diseases. Provide specific hygiene practices that can help reduce the risk of infection. **[10 marks]**

3. (a) Explain why water wet glass but mercury does not. **[8 marks]**

(b) i. What is a satellite? **[3 marks]**

ii. State any four uses of artificial satellite. **[5 marks]**

c) State and explain Newton's laws of motion **[9 marks]**

4. a. State the principle of conservation **[2 marks]**

b. Explain to your learners why Pluto is no longer considered as a planet **[4 marks]**

c. i. Explain what is meant by the scientific term ***eclipse.***

ii. With the aid of an appropriate diagram describe how eclipse of the sun occurs **[6 marks}**

d. i. Without using ***a push or a pull***, how would you explain the concept of force to a Basic Six learner?

ii. If a metallic toy car of mass 10kg is constantly pushed by a child with force 20N, how much acceleration would be given to the car if all opposing forces such as friction are neglected? **[6 marks]**

e. Explain each of the following as it relates to the organization of the Basic School Science Curriculum.

i. Strand (2marks)
ii. Substrand (2 marks)
iii. Content Standard (2 marks)
iv. Indicators (2 marks)
v. Examplars (2 Marks)

5. a. The philosophy of the Ghanaian standard-based science curriculum emphasizes inquiry-based learning and constructivism for effective science teaching and learning. Explain:
   ($\propto$) Inquiry based science teaching
   (β) Constructivism in science teaching

[6 marks]

b. Forces are basically classified into contact and non-contact forces. Name and explain three examples of contact forces. [6 marks]

c. i. What are planets?
   ii. Describe the benefits of the sun to the earth to 5 basic learners. [8 marks]

d. Differentiate between Renewable sources of energy and Non-renewable sources of energy and give one example each. [5 marks]